

Treasures of the Earth

The precious metals and gemstones that have been used for decoration and trade throughout human history have their origin in the rocks that surround us. Many of these began as mineral crystals that formed as a result of geological changes over millennia. The crystals are extracted, then cut, faceted, and polished to be used in jewellery and other decorative items. Organic gems are made of biologically derived matter, such as

pearls produced by oysters, and amber, a form of fossilized tree resin. The financial worth and perceived value of these precious materials can vary from society to society – in some cultures, jade is more valuable than gold, for example. Rubies are among the most highly valued gemstones in the West – the 25.6-carat “Sunrise Ruby”, mounted as a ring by Cartier, was sold for around \$30 million in 2015.

Three rock types

Components of the Earth

There are three major classes of rock: igneous, sedimentary, and metamorphic. Igneous rocks are either formed from magma (molten rock) that has solidified underground, creating intrusive rocks such as granite, or has flowed onto the land or sea bed, forming extrusive rocks such as basalt. Most sedimentary rocks, such as sandstone, are made of deposits laid down on the Earth's surface by wind, water, or ice. Metamorphic rocks are formed when the mineralogical composition of existing rocks is altered. Quartzite, for example, is metamorphosed sandstone.



Igneous rock This example of intrusive igneous rock, granite, is formed inside the Earth when magma cools. Tiny crystals can be seen on its surface.



Sedimentary rock Sandstone usually contains quartz but other minerals can also be present. This example shows patches of iron oxide and flakes of mica.



Metamorphic rock The component minerals of gneiss – mainly quartz and feldspar – tend to separate out into distinct bands of different colours.

Bevelled, tabular crystal, typical of the mineral wulfenite



Minerals

Most gems are cut crystals of minerals. A mineral is defined as a naturally occurring solid with a specific chemical composition and a distinctive crystal structure (see pp.14–15). Each mineral has a unique name based on these two criteria. If either of these changes, it becomes a different mineral with a different name.

Specimen of rough marble

Rocks

The Earth's rocks (see box, left) are made up of naturally occurring aggregates of one or more minerals, although there are a few rocks made from organic substances, such as decayed vegetation, which is the source of coal.

